

CS455: Introduction to Software Engineering

Lecture 7

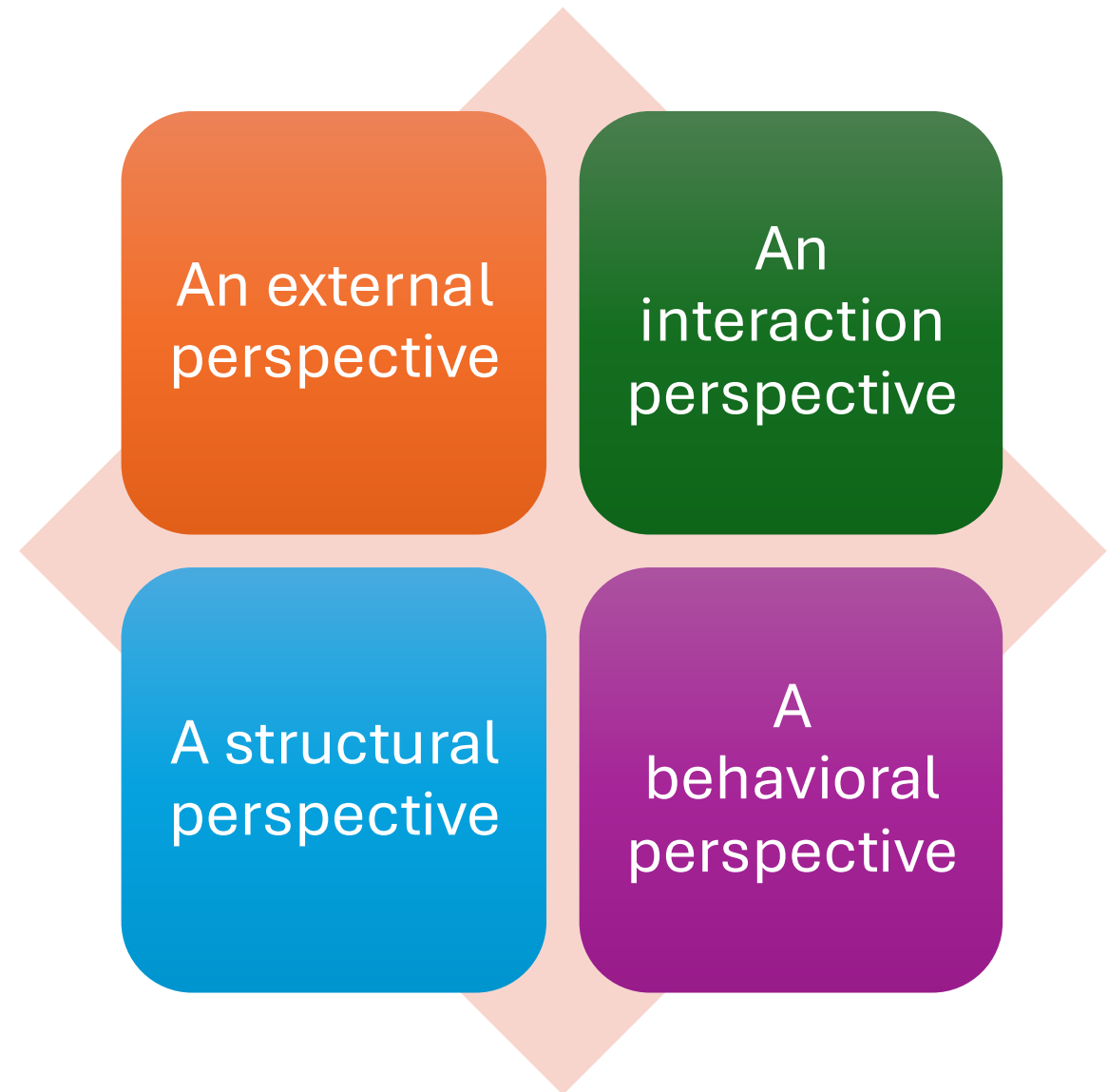
Priyanka Bagade

System Modeling

System Modeling

- Process of developing abstract models of a system, with each model presenting a different view or perspective of that system.
 - Graphical models – e.g. UML
 - Mathematical model
- Models are used during
 - the requirements engineering process to help derive the detailed requirements for a system,
 - the design process to describe the system to engineers implementing the system, and
 - after implementation to document the system's structure and operation.
- Developed for both existing and new systems

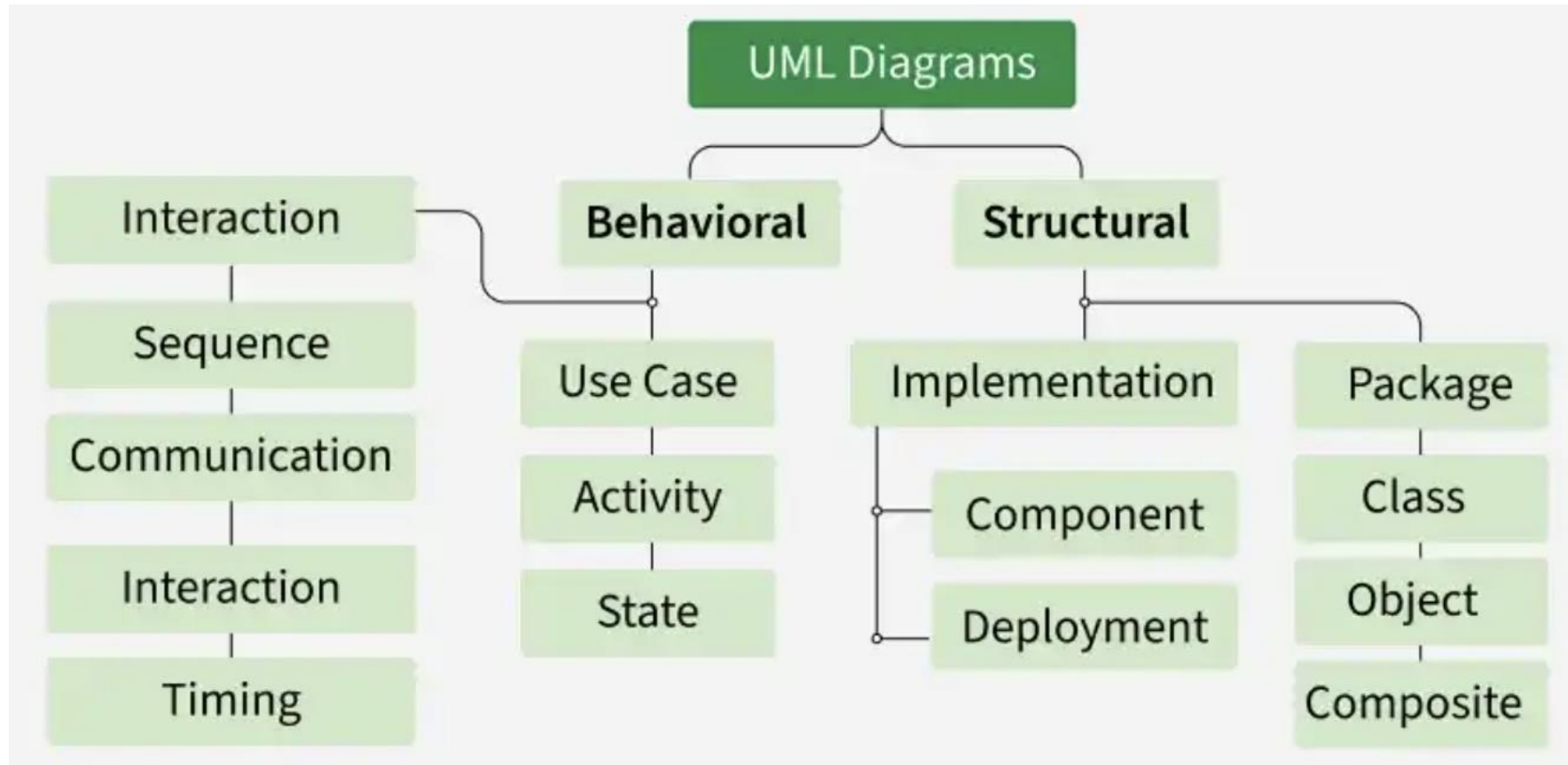
Models from different system perspectives



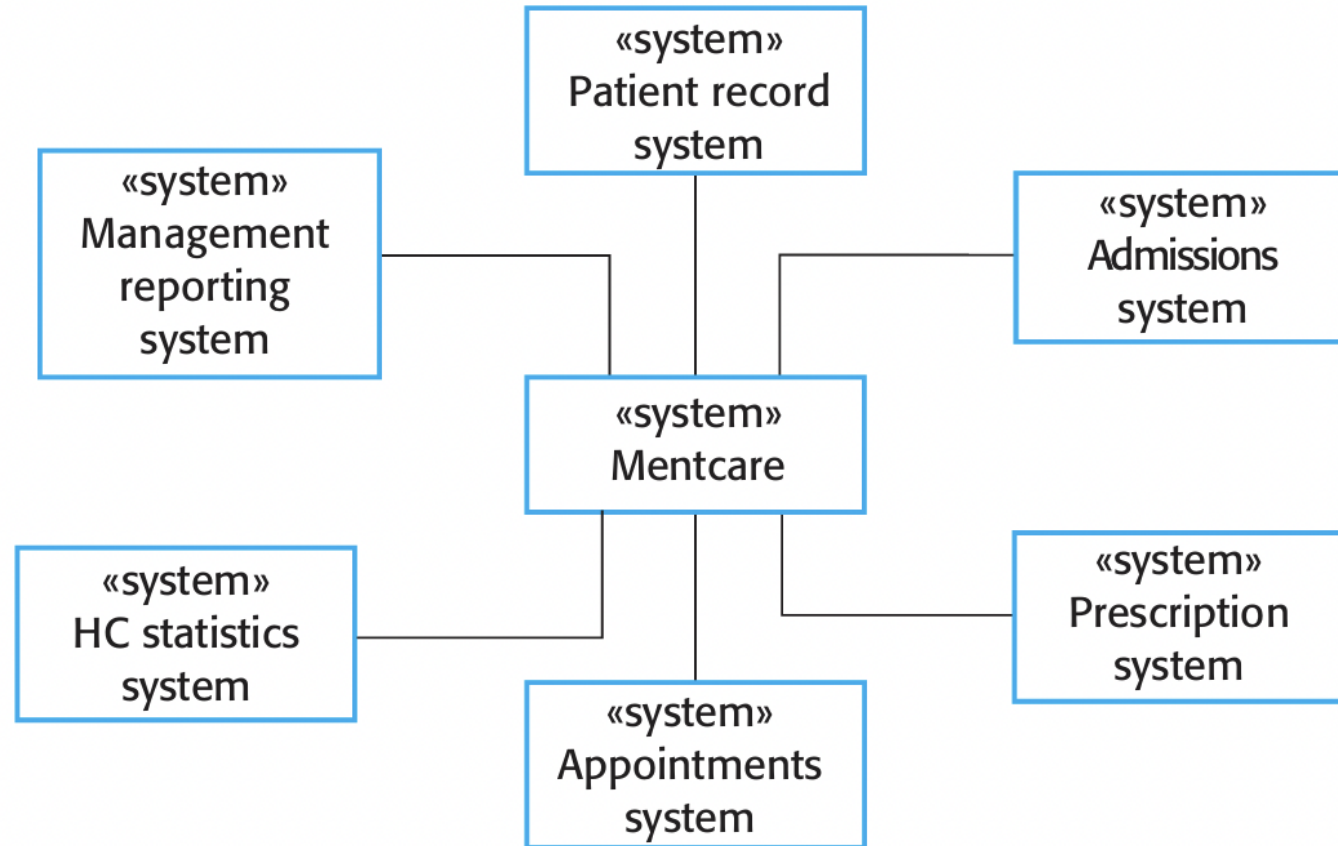
Graphical models

- Use of UML with 13 different diagram types e.g. activity diagram, use case diagram, sequence diagram, class diagram, state diagram
 - As a way to stimulate and focus discussion about an existing or proposed system.
 - As a way of documenting an existing system. When models are used as documentation.
 - As a detailed system description that can be used to generate a system implementation.

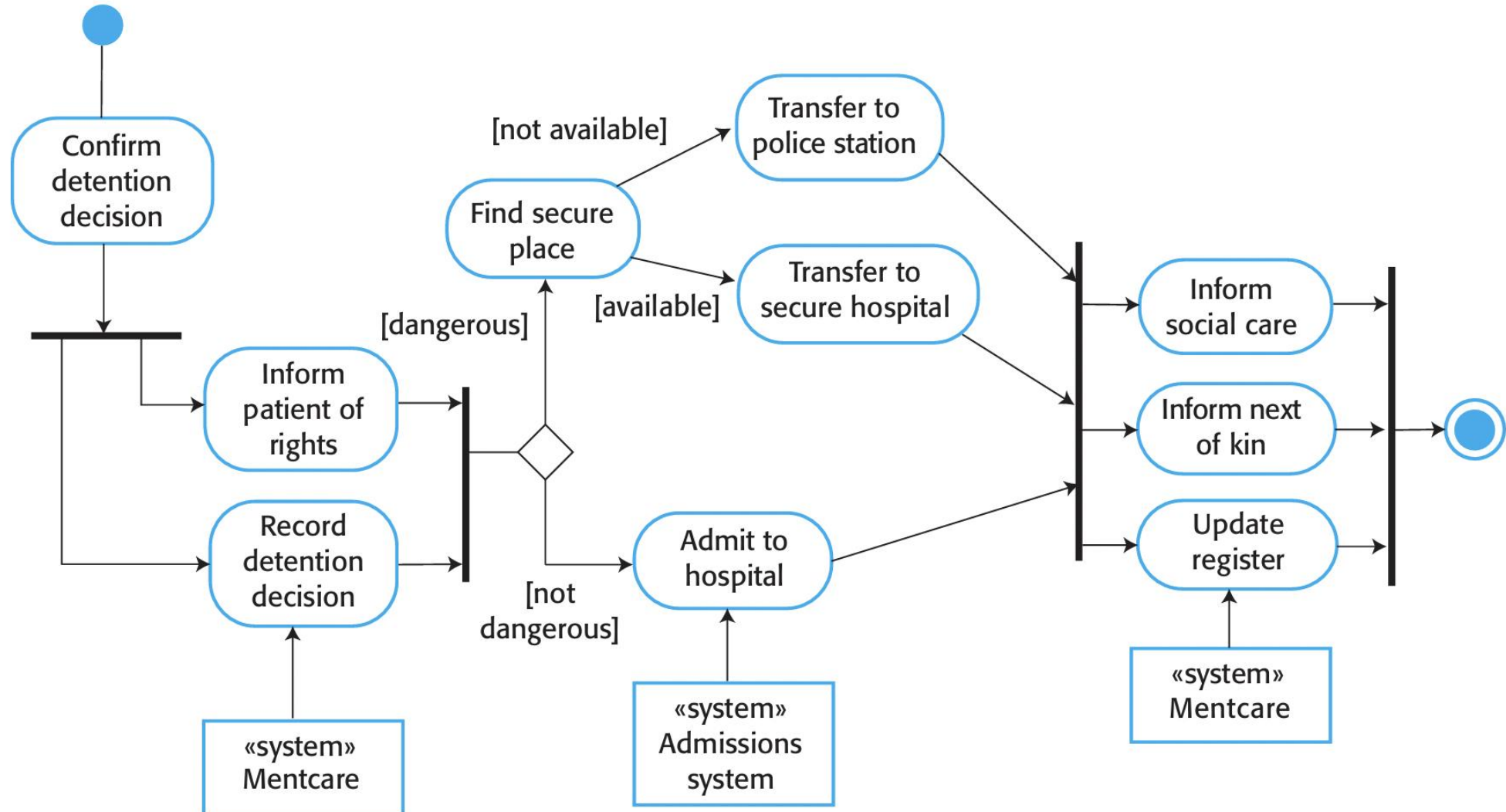
UML Diagrams



Context Models



UML Activity Diagram



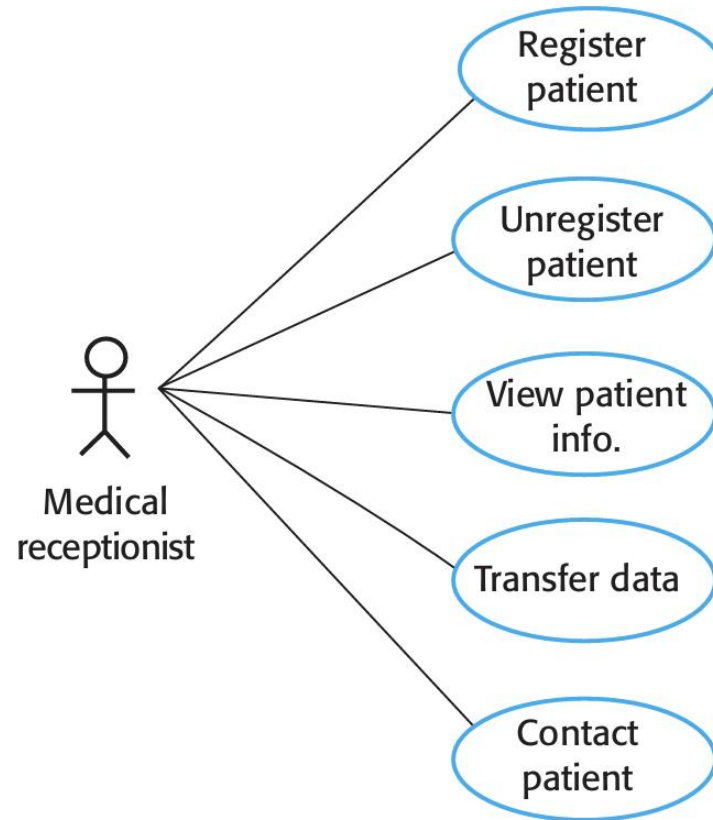
Interaction models

- user inputs and outputs
- interaction between the software being developed and other systems in its environment
- interaction between the components of a software system
- Approaches:
 - Use case modeling - model interactions between a system and external agents (human users or other systems).
 - Sequence diagram - model interactions between system components, although external agents may also be included.

Use Case Modeling

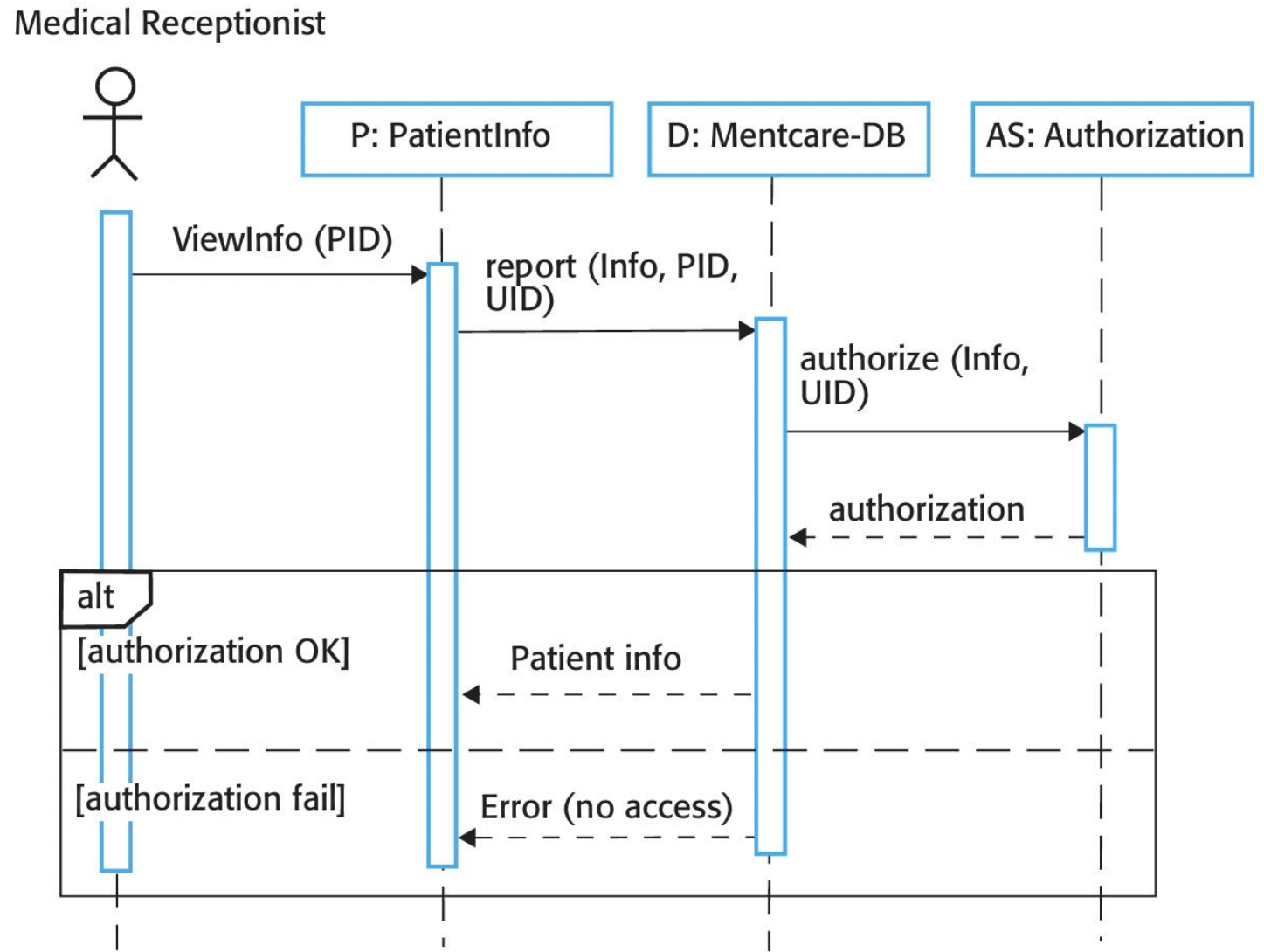
Mentcare system: Transfer data	
Actors	Medical receptionist, Patient records system (PRS)
Description	A receptionist may transfer data from the Mentcare system to a general patient record database that is maintained by a health authority. The information transferred may either be updated personal information (address, phone number, etc.) or a summary of the patient's diagnosis and treatment.
Data	Patient's personal information, treatment summary
Stimulus	User command issued by medical receptionist
Response	Confirmation that PRS has been updated
Comments	The receptionist must have appropriate security permissions to access the patient information and the PRS.

Use case modeling for a specific user

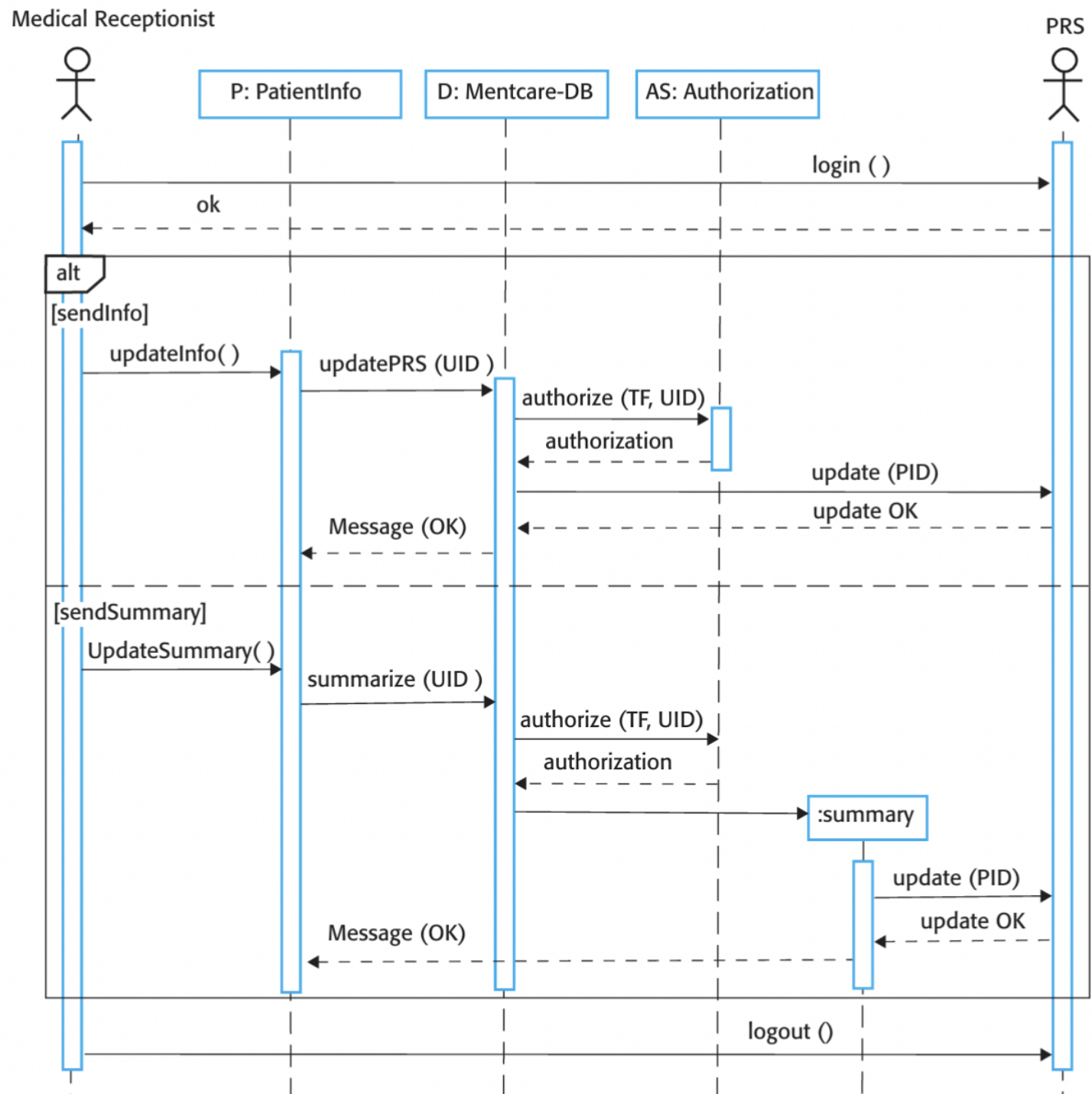


Sequence Diagram

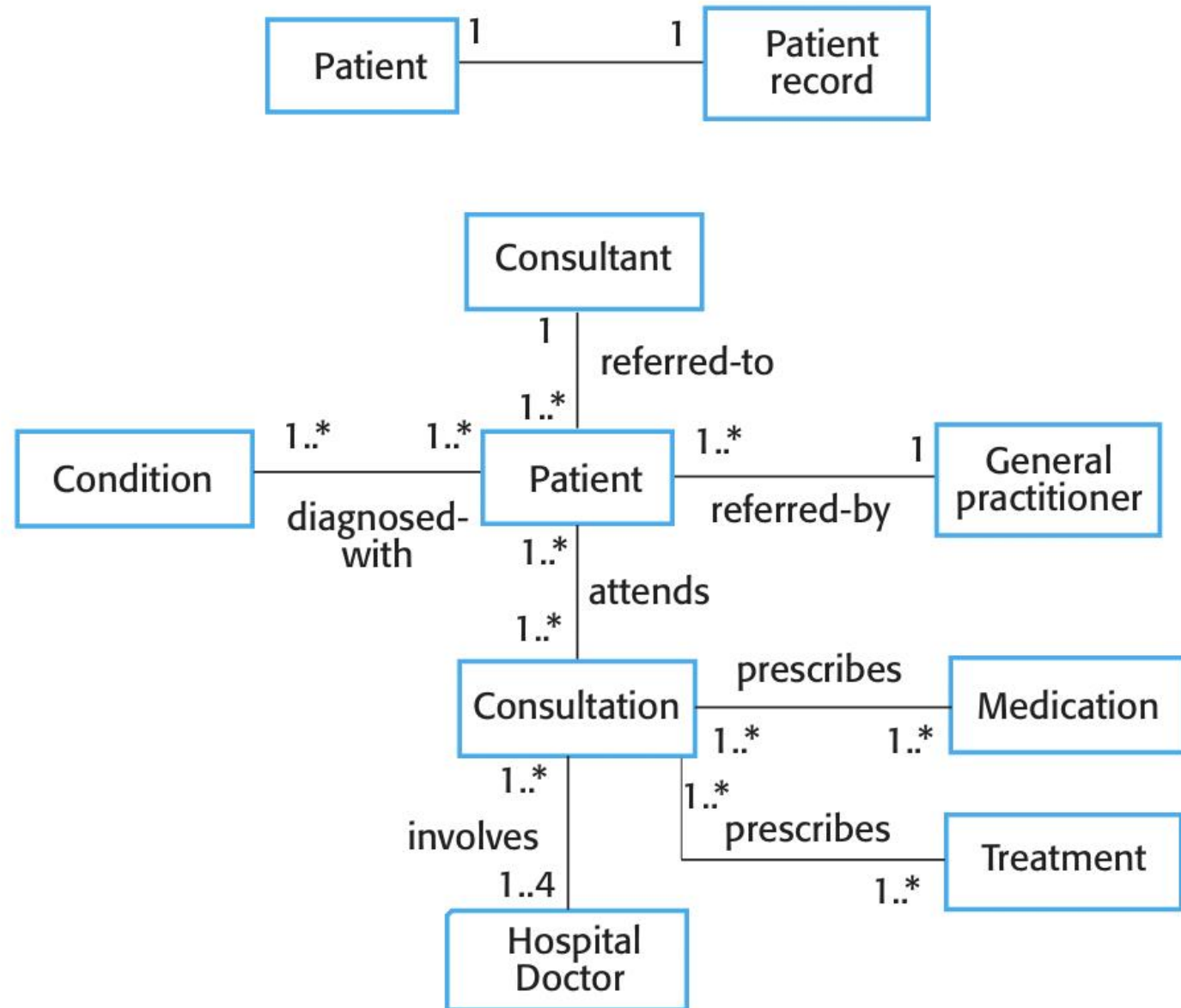
Sequence diagram for
View patient information



Sequence model for Transfer Data



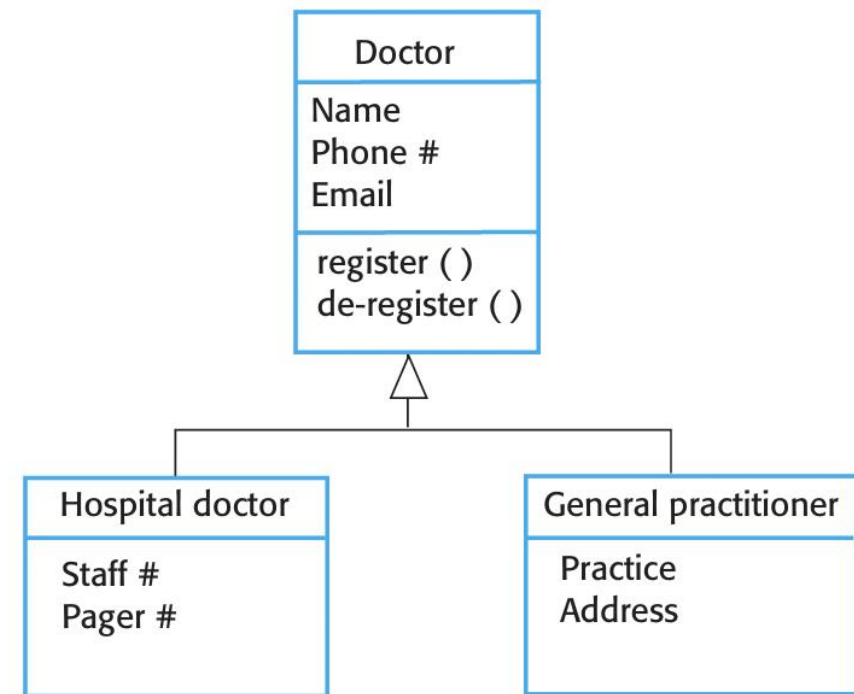
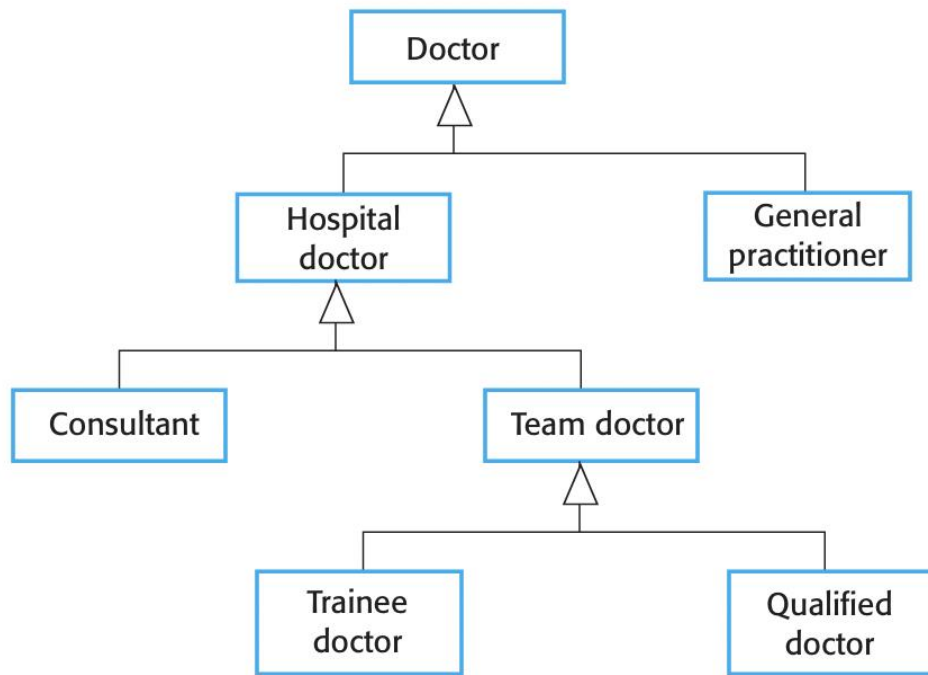
Structure Models – Class Diagram



Consultation Class

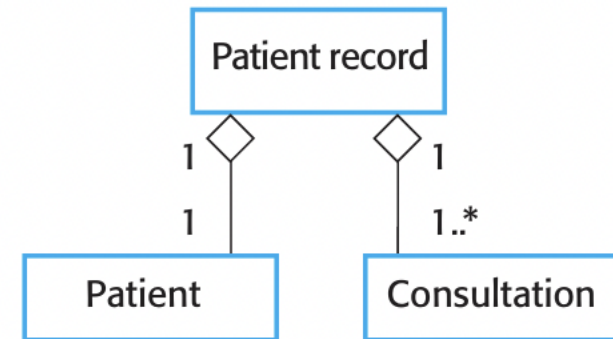
Consultation
Doctors Date Time Clinic Reason Medication prescribed Treatment prescribed Voice notes Transcript ...
New () Prescribe () RecordNotes () Transcribe () ...

Structure Models - Generalization



Structure Models - Aggregation

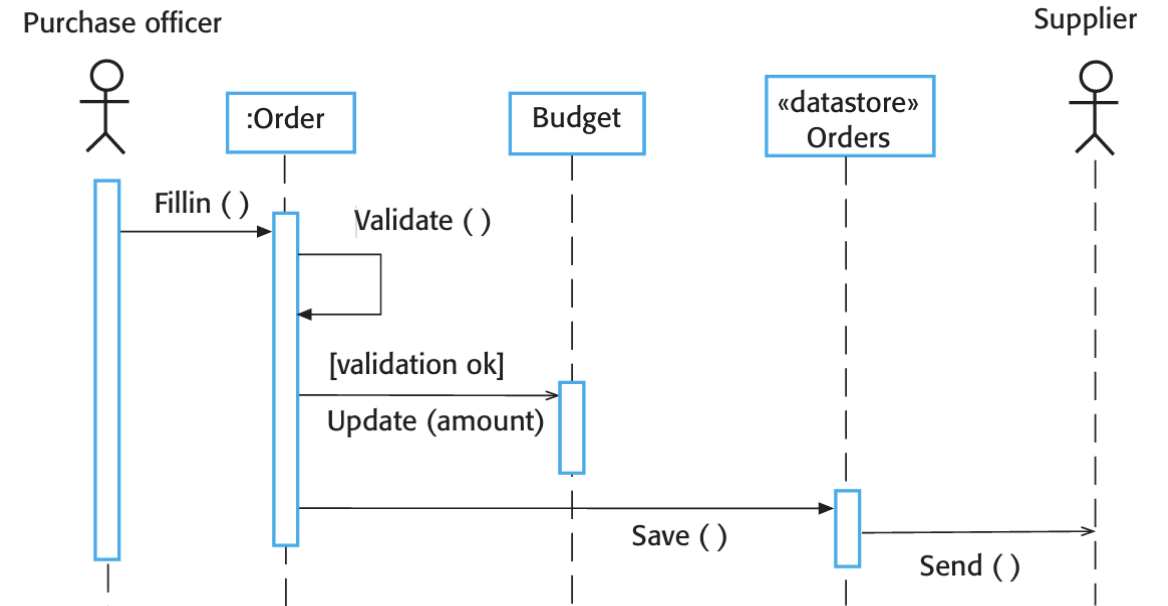
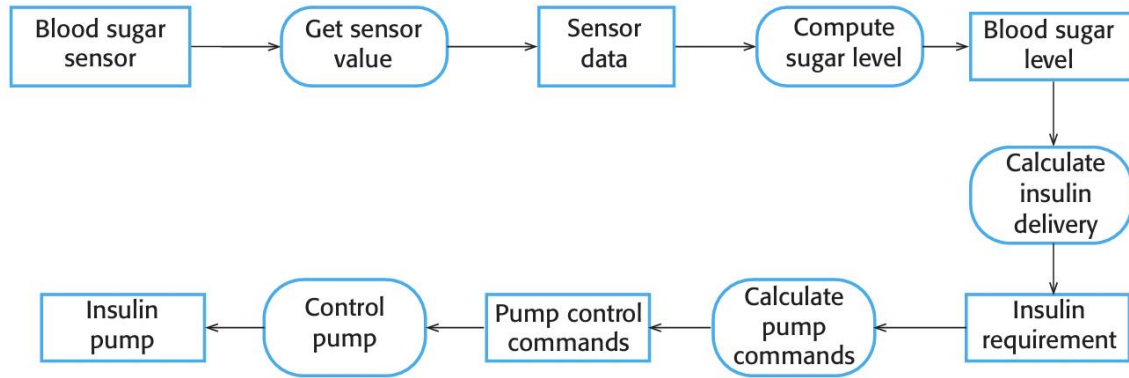
- Objects in the real world are often made up of different parts.
- E.g. a study pack for a course may be composed of a book, PowerPoint slides, quizzes, and recommendations for further reading.



Behavioral Models

- models of the dynamic behavior of a system as it is executing
- System responds to stimuli which can be
 - **Data** becomes available that has to be processed by the system. The availability of the data triggers the processing.
 - An **event** happens that triggers system processing. Events may have associated data, although this is not always the case.

Activity Model



Event Driven Modeling

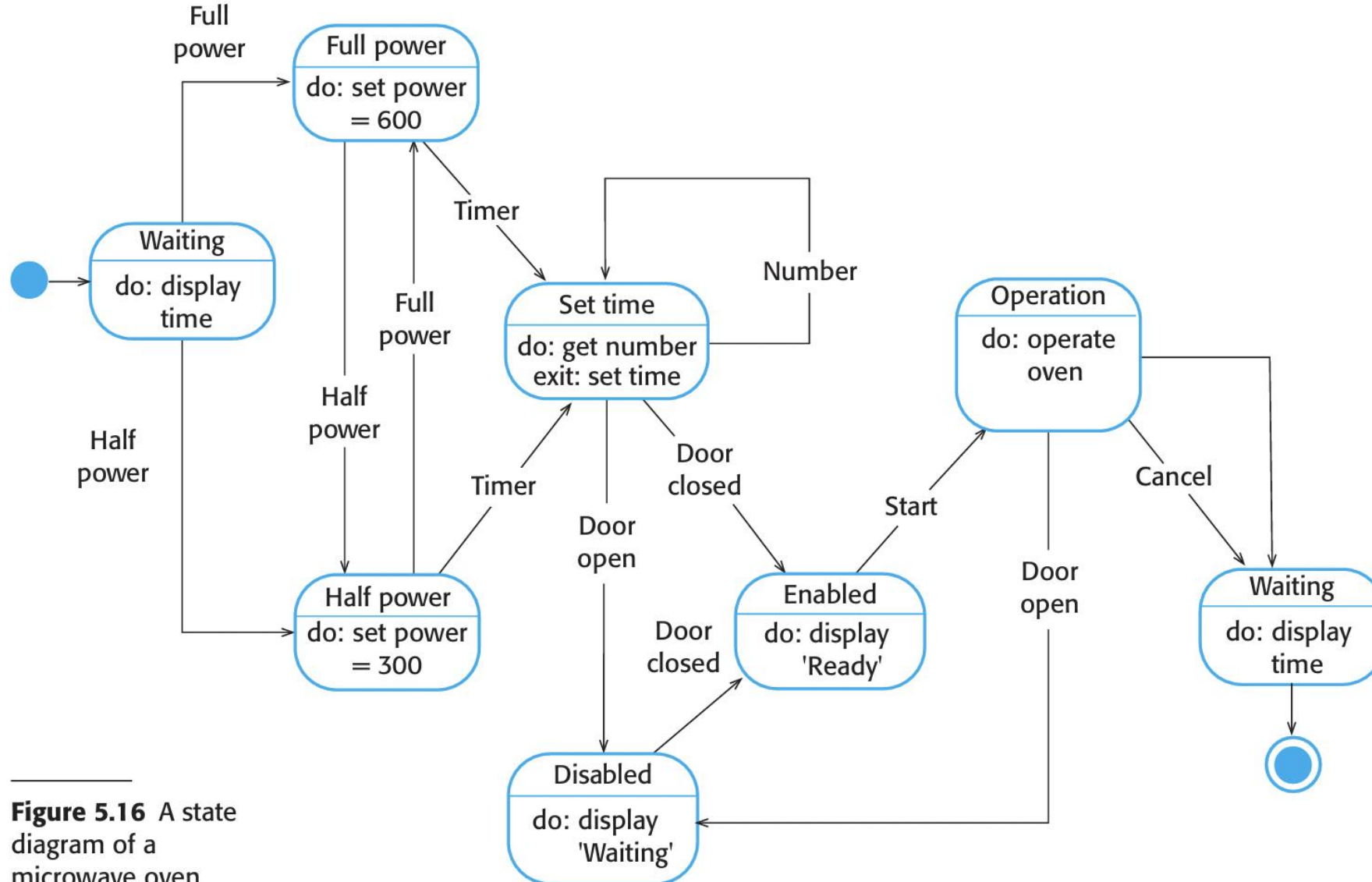


Figure 5.16 A state diagram of a microwave oven

Event Driven Modeling

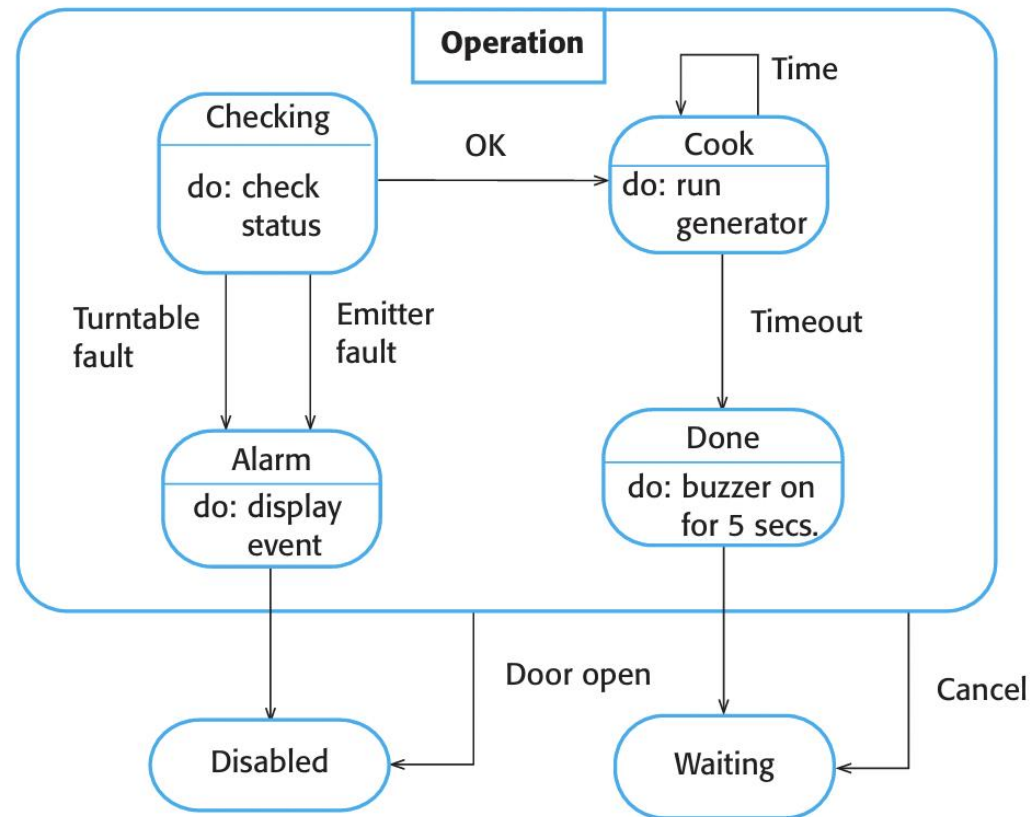
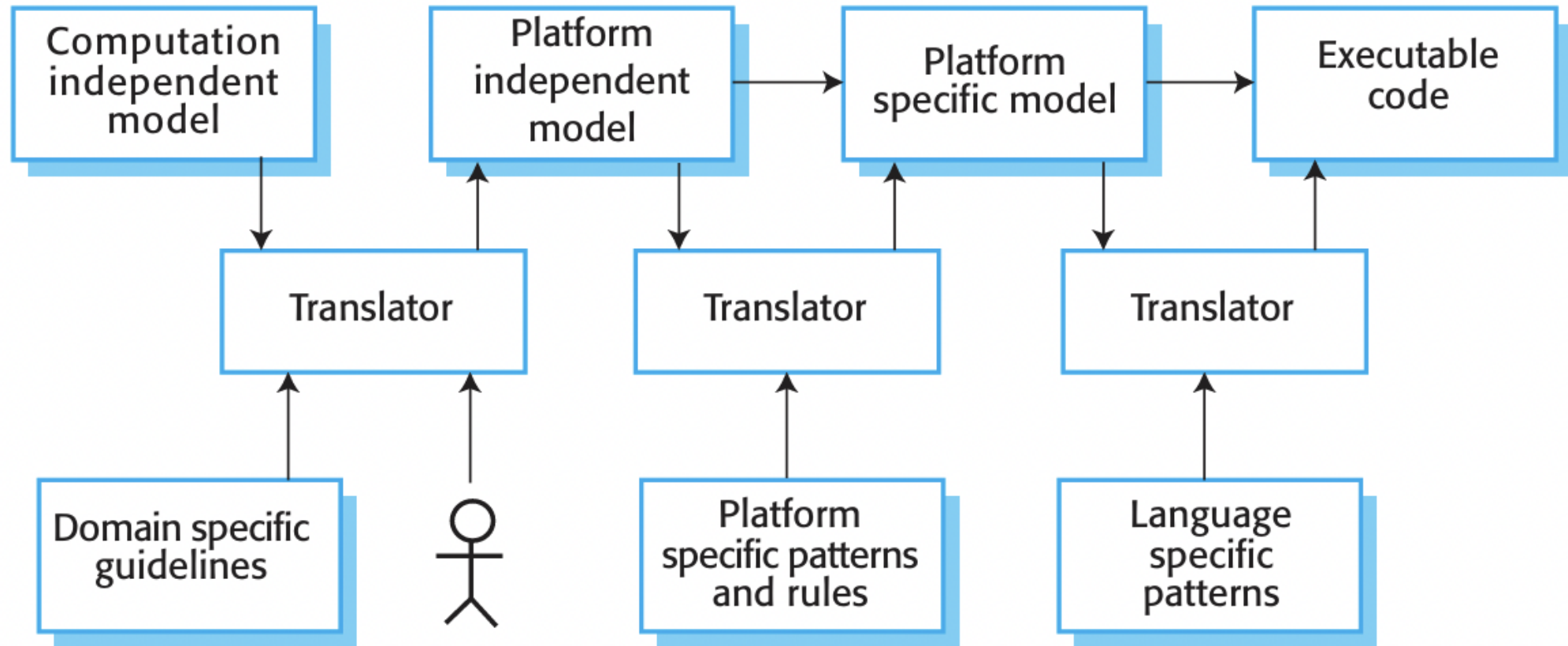


Figure 5.17 A state model of the Operation state

Model Driven Architecture



Model Driven Architecture - platform-specific models

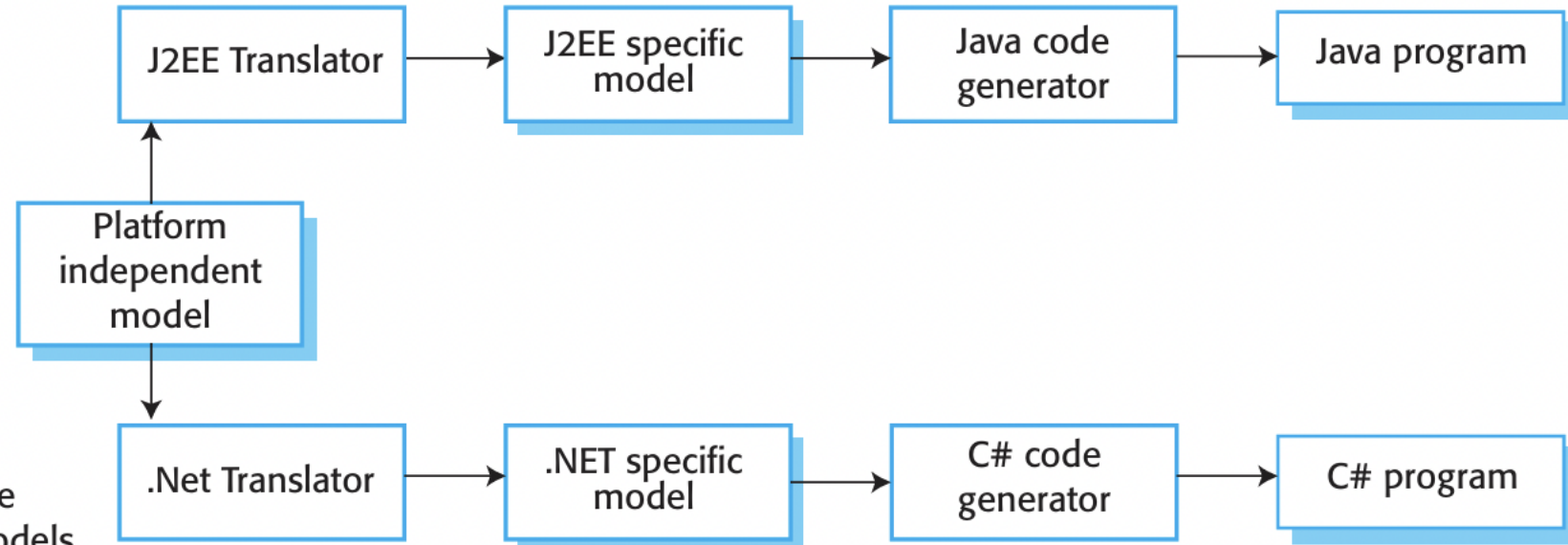


Figure 5.20 Multiple platform-specific models

References

- Chapter 5, Software engineering: [Software Engineering 10e Edition by Ian Sommerville](#)